PROBLEM SET 8

1. Find the simplest possible logic expressions for a 2421-to-51111 BCD code converter.
2. Draw a logic circuit that generates an odd parity bit for the 3321 BCD code.
3. Design the following code converters: (a) 5211 to 2421, (b) 4-bit binary to Excess-3 and (c) 4-bit BCD to Gray.
4. Design a logic circuit to generate an even parity bit for a 3 bit binary input
5. Realize the logic expression given below using a
   (i) 8:1 MUX (ii) 16:1 MUX
   F = $\boldsymbol{\Sigma}$ m(0, 1, 3, 5, 8, 11, 12, 14, 15)
6. Implement the function F(a, b, c, d) = a$\overline{b}$ + acd + b$\overline{c}$ + d using a MUX with 3 select inputs.
7. Implement the following multiple output combinational logic circuit using a 4-line to 16-line decoder.

   $F_1 = \Sigma\, m(0, 1, 4, 7, 12, 14, 15)$
   $F_2 = \Sigma m(1, 3, 6, 9, 12)$
   $F_3 = \Sigma\, m\ (2, 3, 7, 8, 10)$
   $F_4 = \Sigma\, m\ (1, 3, 5)$

8. Each of the eight full-adders in an 8-bit parallel ripple carry adder exhibits the following propagation
   delay:
   A to S and Cout: 20 ns
   B to S and Cout: 20 ns
   Cin to S : 30 ns
   Cin to Cout: 25 ns
   Determine the maximum total time for the addition of two 8-bit numbers.

9. Show the additional logic circuitry necessary to make the 4-bit look-ahead carry adder into a 5-bit adder.

10. You wish to detect only the presence of the codes 1010, 1100, 0001, and 1011. An active- HIGH output is required to indicate their presence. Develop the minimum decoding logic with a single output that will indicate when any one of these codes is on the inputs. For any other code, the output must be LOW.

Additional Problems:

11. Design an voting system for a class of strength 64 that counts yes/no opinion and display the total number of Yes/No.